

ADULT INCOME

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GROUP 1

DATA CLEANING



Train/Test Split

- Split the dataset into 70% training and 15% Validation and 15% testing
- Perform EDA and preprocessing only on the training set
- Keep the test dataset untouched for future model evaluation

Handling Missing Values

- Replace “?” entries with NaN
- Impute categorical values using mode
- Impute numerical values using mean or median

Encoding

- Apply label encoding to binary variables such as gender and income
- Apply one-hot encoding to categorical variables (e.g., workclass, occupation, marital-status)

Feature Engineering

- Create meaningful derived features such as Capital Net, Age_Group, and High hours
- Group or simplify categorical variables if needed.

Scaling and normalization

- Apply MinMaxScaler to numerical features such as age, hours-per-week, capital-gain, and capital-loss to normalize skewed distributions

CLASSIFICATION MODEL

Best Model
Gradient Boosting
Classifier

Best Overall Model
Bayesian Stacking

Worst Model
SVC

Classification Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.84	0.71	0.55	0.62	0.89
Decision Tree Classifier	0.85	0.68	0.67	0.68	0.89
Random Forest Classifier	0.84	0.7	0.6	0.65	0.89
Gradient Boosting Classifier	0.86	0.76	0.6	0.67	0.92
K-Nearest Neighbors (KNN)	0.84	0.66	0.64	0.65	0.87
Support Vector Classifier (SVC)	0.8	0.72	0.28	0.4	0.84
Voting (Soft) - Top 3 Models	0.8633	-	-	-	0.92
Ensemble (Bayesian Stacking)	0.8682	-	-	-	-



THE OUTCOME

- The model can distinguish high earners and average earners based on their demographics with 86.82% accuracy
- Marital Status (Married-civ-spouse) and Education Level (Educational-num) are the variables which have the highest correlation with income
- The model is conservative. It has high Precision (0.76) but a moderate Recall (0.60)

REAL LIFE APPLICATIONS

- Target high-value customers for premium credit cards
- Estimate the economic health of a specific demographic
- Identify which life factors (like gaining more education) have the most significant impact on an individual's earning potential.